Tony DeFazio

Associate Research Scientist
defazio@umich.edu
Public copy: student names are withheld and shown as counts. Full CV on request — tony@tonydefazio.com

Education and Training

Education

08/1985-06/1989	BS, Physics, Case Western Reserve University, Cleveland, OH, United States
08/1989-06/1996	PhD, Biology, University of Southern California, Los Angeles, CA, John P. Walsh, PhD

Postdoctoral Training

06/1996-02/1997	Postdoctoral Research, Mental Retardation Research Center, Lab of Michael S. Levine, University of California, Los Angeles, Los Angeles, CA, Michael S. Levine
02/1997-06/2000	Postdoctoral Research, Neurobiology, Lab of John J. Hablitz, University of Alabama, Birmingham, Birmingham, Al, John J. Hablitz

Work Experience

Academic Appointment

06/2000-06/2002	Research Assistant Professor, University of Virginia, Charlottesville
08/2003-08/2004	Associate Scientist, University of Miami, Miami
08/2004-10/2005	Research Assistant Professor, Physiology Department, University of Miami, Miami
10/2005-05/2010	Research Assistant Professor, Neurology Department, University of Miami, Miami
05/2010-12/2011	Assistant Professor, University of Miami, Miami
01/2012-09/2018	Assistant Research Scientist, Molecular & Integrative Physiology Department, University of Michigan, Ann Arbor
09/2018-Present	Associate Research Scientist in Molecular and Integrative Physiology, University of Michigan - Ann Arbor, Ann Arbor

Industry

07/2002-08/2003	Scientist, Johnson & Johnson Pharmaceutical, San Diego

Research Interests

- Excitation-release coupling is a common feature in physiology. Action potentials depolarize the membrane potential near the release site, open voltage-gated calcium channels, and calcium influx triggers release. Pulsatile release of gonadotropin releasing hormone (GnRH) from neurons projecting to the median eminence regulates the release of luteinizing hormone and follicle stimulating hormone, which in turn drive gonadal function. Kisspeptin is a neuropeptide originating in two hypothalamic nuclei (arcuate and AVPV) and is a potent activator of GnRH neurons. Building on a decade of electrophysiology studies in GnRH and arcuate kisspeptin neurons, I am now using transgenic encoded calcium indicators to study a new, parallel axis of activity in these neurons. I demonstrated two classes of calcium events and their differential response to the neurokinin-3 receptor agonist senktide (paper in press). Unpublished work extends this to action potential-independent calcium signaling and to coordination that differs between the two classes of events.

- Dynamic clamp permits direct evaluation of the realtime interaction between simulated conductances (voltage-gated and ligand-gated ion channels) and the electrophysiologic properties of neurons. Dynamic clamp provides a link between computational modeling and the temporal relationship between voltage-gated ion channels and synaptic potentials. Using classic whole cell electrophysiology, dynamic clamp, and in silico modeling, we have previously demonstrated that voltage-gated potassium channels can influence the amplitude of depolarizing GABA postsynaptic potentials in arcuate kisspeptin neurons. A similar role is hypothesized in GnRH neurons which have a stronger depolarizing GABA response. Dynamic clamp and in silico modeling can help elucidate the role of ion channels in the release of kisspeptin and GnRH, and in the response of GnRH neurons and gonadotropes. This work preceded my transition to agentic coding tools.

Research Computing and AI-Assisted Development

- I design and maintain calcium-imaging analysis infrastructure end to end. Proprietary-format imaging series of up to 44,000 frames are ingested through Bio-Formats. Motion correction is processed as Slurm array jobs on the U-M Great Lakes cluster, with the job manifest derived deterministically from the input folder. Residual motion-correction drift is scored by a logged quantitative metric rather than by visual inspection, and baseline estimation keeps the slow fluorescence drift as a signal. Events are detected with a custom routine that does not depend on spike inferences. Custom summary figures are built in MATLAB and R.

- I have built several open-source tools (tonydefazio.com): Draughtsman traces the architecture of PyTorch deep learning models, with nine reference models written out in full behind a coverage gate that fails unless every compute operation lands in exactly one stage (PyPI: draughtsman-nn; DOI 10.5281/zenodo.22286341). Bugarach detects coordinated events in multicell recordings like calcium imaging. The detectors are optimized and benchmarked against synthetic data with planted coordinated events using parameters derived from the dataset. Murderboard is an adversarial AI review harness for written deliverables, running eleven role-specialized passes over a document (DOI 10.5281/zenodo.22063835). Each of these is a public-facing website and repository (accessible via tonydefazio.com and github.com/syncytium2).

- This work was greatly accelerated by coding agents under robust checks and reviews. Custom built pre- and post-tool gates block known agent failure modes and continuously evolve as agent behavior changes. The gates are mutation-tested: each tool is broken on purpose and its self-test is required to go red, so a check that cannot fail is caught before it is trusted. The boundary between hand-written and agent-directed work is April 2026; my MATLAB codebase contained ~2,000 manually coded files in the imaging pipeline prior to the first agent commit. I now routinely code in R, Svelte/Vite, Python, and whatever programming environment and tech stack is most appropriate for the task.

Grants

All grants listed below are awarded to Suzanne M. Moenter, who is the Principal Investigator. The role shown is mine. These are not my awards and not my funds.

Current Grants

R01HD041469:*Central Actions of Estrogens: Effects on GnRH Neurons:*
Co-I (Principal Investigator:Sue Moenter)
NIH-DHHS-US
09/2023 - 05/2028

R37HD034860:*Cellular and molecular bases for rhythmic GnRH release:*
Co-I (Principal Investigator:Sue Moenter)
NIH-DHHS-US
06/2022 - 05/2027

R01HD104345:*Development of the GnRH neuronal network and effects of prenatal androgen exposure:*
Co-I (Principal Investigator:Sue Moenter)
NIH-DHHS-US
05/2021 - 01/2027

Past Grants

R01HD041469:*Central Actions of Estrogens: Effects on GnRH Neurons:*
Co-I (Principal Investigator:Sue Moenter)
NIH-DHHS-US
09/2018 - 08/2023

R37HD034860:*Cellular and molecular bases for rhythmic GnRH release:*
Co-I (Principal Investigator:Sue Moenter)
NIH-DHHS-US
09/2017 - 05/2022

U54HD028934:*Development of the GnRH neuronal network and effects of prenatal androgens:*
Co-I (Principal Investigator:Sue Moenter)
NIH-DHHS-US-SubK sourced funding through University of Virginia
04/2014 - 03/2020

R01HD041469:*Central Actions of Estrogens: Effects on GnRH Neurons:*
Co-I (Principal Investigator:Sue Moenter)
NIH-DHHS-US
04/2013 - 08/2018

R01HD034860:*Cellular and Molecular Bases for Rhythmic GnRH Release:*
Co-I (Principal Investigator:Sue Moenter)
NIH-DHHS-US
07/2012 - 08/2017

0749840:*Luminescent Switches for Fluorescence Nanoscopy:*
Funded by
Directorate for Mathematical & Physical Sciences
02/2008 - 01/2012

Honors and Awards

National

1999 - 2000	AES/Epilepsy Foundation Research Training Fellowship
2000	American Epilepsy Society Junior Investigator Travel Award
2001	Astra-Zeneca Endocrine Society Travel Award

Institutional

2023	Endowment for Basic Science Teaching Award, University of Michigan, Molecular & Integrative Physiology, United States, The EBS Teaching Award was established in 2011 to recognize faculty in units affiliated with the Endowment for the Basic Sciences

Study Sections, Editorial Boards, Journal & Abstract Review

Study Sections

National

2007	NIH Study section: ZNS1 SRB-P, (Ad Hoc)
2009	American Heart Association, Basic Cell and Molecular Biology, Region II, (Ad Hoc)

Editorial Boards / Journal & Abstract Reviews

2017 – Present	Reviewer Journal of Neurochemistry
2017 – Present	Reviewer Journal of Scientific Reports
2018 – Present	Reviewer Journal of Neuroscience
2018 – Present	Reviewer Frontiers in Pharmacology
2018 – Present	Reviewer Frontiers in Neuroscience
2018 – Present	Reviewer Frontiers in Endocrinology
2018 – Present	Reviewer Frontiers in Cellular Neuroscience
2026	Reviewer, Stroke

Editorial Boards

2022 - Present	Associate Editor, Cellular Neurophysiology, Frontiers in Cellular Neuroscience

Teaching

Mentorship

Graduate Student

2009 - 2018	7 graduate students mentored - University of Michigan (4), University of Miami (1, MD/PhD), Florida International University (1), University College Dublin (1). Names withheld on this public copy; full CV on request.

Undergraduate Student

2009 - 2011	6 undergraduate students mentored - University of Miami (5), Miami-Dade Community College (1). Names withheld on this public copy; full CV on request.

Teaching Activity

Institutional

01/1990 - 01/1991	USC Teaching assistant, General Physiology Laboratory (BIOL 301) and Neurobiology Laboratory (BIOL 422)
01/1993 - 01/1995	USC Teaching assistant, Biology of Aging and Development (GERO 210)
01/1995 - 01/1996	Cal State LA Instructor, Biology of Human Aging (BIOL 484N)
01/1996	Cal State LA Co-Instructor, Natural History of Animals (BIOL 155)
01/2001	University of Virginia Lecture on Hypothalamus, Behavioral and Cognitive Neuroscience (NESC 720)
01/2009	UMiami Clinical Neurophysiology for Neurology Residents and Fellows Course director, Dr. Cynthia Harden Basic science lectures I & II (Translational Neuroscience and Channelopathies)
01/2010	UMiami Neuroscience Journal Club NEU600, course leader Dr. Ian Hentall Faculty leader (4-5 week segment) Topic: The neurovascular unit
01/2011	UMiami Guest lecturer, BIL299. Dr. Akira Chiba, course leader "This will be the first of a series of 'exclusive interviews with real-life scientists'. It has two parts, about 30 minutes each: personal interview (a Larry King style) and research talk (a condensed version for the uninitiated)."
01/2012	UMich PIBS 503 Responsible Conduct of Research Course (2012-present). I teach the Fraud, Fabrication, and Plagiarism section. Dr. Kate Barald, course director
01/2013	UMich Physiol 406/506 Extreme physiology (Fall 2013). Delivered two lectures on hibernation and taste/olfaction. Dr. Thomas Michael Ferrari, course director
01/2013	UMich Physiol 505 Advanced human physiology laboratory (Fall 2013). Led discussion on action potential mechanisms and assisted with applied laboratory electrophysiological recordings from leach neurons. Dr. Timothy Houchin, course director
01/2015 - 2022	UMich Physiol 576 Signal Transduction: Ligand and voltage-gated channels. 2 x 45 minute lectures
01/2015 - Present	UMich Physiol 502 Human Physiology: Reproduction. 3 x 1.5 hour lectures

01/2015 - 01/2018	UMich Physiol 592 Integrative Neuroscience: Neurotransmitters and receptors. 3 x 1.5 hour lectures
01/2017 - 01/2019	UMich Physiol 577 Membrane and Cell Physiology: GABAA receptor signaling. 1 x 45 minute lecture
01/2018 - Present	UMich Physiol 592 Integrative Neuroscience: Taste and Olfaction 3 x 1.5 hour lectures
08/2022 - Present	Course Director. UMich Physiol 578: Cellular physiology: key concepts and applications in physiology. Combined and revised material from previous courses (576 and 577). Dynamically responded to student feedback. Lectures range from 9-14 x 50 minute lectures per semester; academic year 2026-27 is 18 x 50 minute lectures (15 fall, 3 spring), University of Michigan, Molecular and Integrative Physiology

Dissertation Committees

2008 - Present	2 dissertation committees, Committee Member - University of Miami, Department of Chemistry (2). Names and dissertation titles withheld on this public copy; full CV on request.

Memberships in Professional Societies

1990 - Present	Society for Neuroscience
1999 - 2003	American Epilepsy Society
2003 - 2005	Association for Chemoreception Sciences
2006 - 2008	American Heart Association
2018 - Present	Pan American Neuroendocrine Society

Scholarly Activities

Presentations

Extramural Invited Presentation

1. "Steroid regulation of GnRH neurons.", Center for Research in Reproduction, University of Virginia, 12/2006, Virginia

Other

1. Ischemic preconditioning increases GABA synaptic inputs to CA1 hippocampal neurons, Department of Neurology Grand Rounds, University of Miami, 01/2007, Coral Gables, FL
2. A biosensor for early detection of melanoma, Department of Dermatology Grand Rounds, University of Miami., 06/2007, Coral Gables, FL
3. Persistent interneuron dysfunction in rat hippocampus after cardiac arrest, Department of Neurology Grand Rounds, University of Miami., 10/2007, Coral Gables, FL
4. Progress towards a biosensor for melanoma, Department of Dermatology Grand Rounds, University of Miami., 05/2008, Coral Gables, FL
5. Diverse interneuron adaptations after ischemia: a role for GABA in neuroprotection and neuronal dysfunction., Neuroscience Institute, Morehouse School of Medicine., 10/2008, Atlanta, GA
6. Diverse interneuron adaptations after ischemia, Department of Neurology Grand Rounds. University of Miami., 11/2008, Coral Gables, FL
7. Diverse roles for GABAergic interneurons after cerebral ischemia, Department of Biology seminar. Florida Atlantic University., 03/2009, Boca Raton, FL
8. HIV Transactivator Protein Tat Modulates Synaptic Transmission in the Mouse Hippocampus: Implications for Cognitive Decline and NeuroAIDS, Department of Microbiology Grand Rounds, University of Miami, 04/2010, Coral Gables, FL
9. Summer Scholars: NeuroAIDS update, lecture to high school students visiting University of Miami under the Summer Scholars Program, 07/2010, Coral Gables, FL
10. Neuroprotection via collateral flow after middle cerebral artery occlusion, Department of Neurology Grand Rounds, University of Miami, 12/2010, Coral Gables, FL
11. Summer Scholars: Brain + HIV = NeuroAIDS, lecture to high school students visiting University of Miami under the Summer Scholars Program, 07/2011, Coral Gables, FL
12. Towards a vascular cure for stroke: Cerebral arteries, veins, and ischemia, St. Thomas University., 11/2012, Miami, Florida

Publications/Scholarship

(Co-First Author *; Corresponding author **; Co-Last author ***)

Peer Reviewed

Journal Article

1. **DeFazio T**, Walsh JP: 'Intact' dopaminergic midbrain neurons of the rat display unclamped dendritic Ca2+ currents. *Neurosci Lett*.208(1): 29-32, 04/1996.PM8731167
2. Dunia R, Buckwalter G, **Defazio T**, Villar FD, McNeill TH, Walsh JP: Decreased duration of Ca(2+)-mediated plateau potentials in striatal neurons from aged rats. *J Neurophysiol*.76(4): 2353-2363, 10/1996.PM8899609
3. **Defazio T**, Hablitz JJ: Zinc and zolpidem modulate mIPSCs in rat neocortical pyramidal neurons. *J Neurophysiol*.80(4): 1670-1677, 10/1998.PM9772230

4. **Defazio RA**, Hablitz JJ: Reduction of zolpidem sensitivity in a freeze lesion model of neocortical dysgenesis. *J Neurophysiol*.81(1): 404-407, 01/1999.PM9914301

5. **DeFazio RA**, Hablitz JJ: Alterations in NMDA receptors in a rat model of cortical dysplasia. *J Neurophysiol*.83(1): 315-321, 01/2000.PM10634874

6. **DeFazio RA**, Pong K, Knusel B, Walsh JP: Neurotrophin-4/5 promotes dendritic outgrowth and calcium currents in cultured mesencephalic dopamine neurons. *Neuroscience*.99(2): 297-304, 01/2000.PM10938435

7. **DeFazio RA**, Keros S, Quick MW, Hablitz JJ: Potassium-coupled chloride cotransport controls intracellular chloride in rat neocortical pyramidal neurons. *J Neurosci*.20(21): 8069-8076, 11/2000. PM11050128

8. **DeFazio RA**, Hablitz JJ: Chloride accumulation and depletion during GABA(A) receptor activation in neocortex. *Neuroreport*.12(11): 2537-2541, 08/2001.PM11496144

9. Nunemaker CS, **DeFazio RA**, Geusz ME, Herzog ED, Pitts GR, Moenter SM: Long-term recordings of networks of immortalized GnRH neurons reveal episodic patterns of electrical activity. *J Neurophysiol*.86(1): 86-93, 07/2001. PM11431490

10. **DeFazio RA**, Moenter SM: Estradiol feedback alters potassium currents and firing properties of gonadotropin-releasing hormone neurons. *Mol Endocrinol*.16(10): 2255-2265, 10/2002.PM12351691

11. **DeFazio RA**, Heger S, Ojeda SR, Moenter SM: Activation of A-type gamma-aminobutyric acid receptors excites gonadotropin-releasing hormone neurons. *Mol Endocrinol*.16(12): 2872-2891, 12/2002.PM12456806

12. Nunemaker CS, **DeFazio RA**, Moenter SM: Estradiol-sensitive afferents modulate long-term episodic firing patterns of GnRH neurons. *Endocrinology*.143(6): 2284-2292, 06/2002.PM12021193

13. Nunemaker CS, **DeFazio RA**, Moenter SM: A targeted extracellular approach for recording long-term firing patterns of excitable cells: a practical guide. *Biol Proced Online*.5: 53-62, 01/2003.PM12734556

14. Nunemaker CS, Straume M, **DeFazio RA**, Moenter SM: Gonadotropin-releasing hormone neurons generate interacting rhythms in multiple time domains. *Endocrinology*.144(3): 823-831, 03/2003.PM12586758

15. Sullivan SD, **DeFazio RA**, Moenter SM: Metabolic regulation of fertility through presynaptic and postsynaptic signaling to gonadotropin-releasing hormone neurons. *J Neurosci*.23(24): 8578-8585, 09/2003.PM13679427

16. Nunemaker CS, **DeFazio RA**, Moenter SM: Calcium current subtypes in GnRH neurons. *Biol Reprod*.69(6): 1914-1922, 12/2003.PM12904316

17. Liu JC, **DeFazio RA**, Espinosa-Jeffrey A, Cepeda C, de Vellis J, Levine MS: Calcium modulates dopamine potentiation of N-methyl-D-aspartate responses: electrophysiological and imaging evidence. *J Neurosci Res*.76(3): 315-322, 05/2004.PM15079860

18. Moenter SM, **DeFazio RA**: Endogenous gamma-aminobutyric acid can excite gonadotropin-releasing hormone neurons. *Endocrinology*.146(12): 5374-5379, 12/2005.PM16123153

19. **DeFazio RA**, Hablitz JJ: Horizontal spread of activity in neocortical inhibitory networks. *Brain Res Dev Brain Res*.157(1): 83-92, 06/2005.PM15939088

20. **DeFazio RA**, Dvoryanchikov G, Maruyama Y, Kim JW, Pereira E, Roper SD, Chaudhari N: Separate populations of receptor cells and presynaptic cells in mouse taste buds. *J Neurosci*.26(15): 3971-3980, 04/2006.PM16611813

21. Kim E, Raval AP, **Defazio RA**, Perez-Pinzon MA: Ischemic preconditioning via epsilon protein kinase C activation requires cyclooxygenase-2 activation in vitro. *Neuroscience*.145(3): 931-941, 03/2007.PM17307294

22. Raval AP, Dave KR, **DeFazio RA**, Perez-Pinzon MA: epsilonPKC phosphorylates the mitochondrial K(+) (ATP) channel during induction of ischemic preconditioning in the rat hippocampus. *Brain Res*.1184: 345-353, 12/2007.PM17988655

23. Park H-P, Nimmagadda A, **DeFazio RA**, Busto R, Prado R, Ginsberg MD: Albumin therapy augments the effect of thrombolysis on local vascular dynamics in a rat model of arteriolar thrombosis: a two-photon laser-scanning microscopy study. *Stroke*.39(5): 1556-1562, 05/2008.PM18323493

24. Dave KR, **DeFazio RA**, Raval AP, Torraco A, Saul I, Barrientos A, Perez-Pinzon MA: Ischemic preconditioning targets the respiration of synaptic mitochondria via protein kinase C epsilon. *J Neurosci*.28(16): 4172-4182, 04/2008.PM18417696

25. **DeFazio RA**, Raval AP, Lin HW, Dave KR, Della-Morte D, Perez-Pinzon MA: GABA synapses mediate neuroprotection after ischemic and epsilonPKC preconditioning in rat hippocampal slice cultures. *J Cereb Blood Flow Metab*.29(2): 375-384, 02/2009.PM18957990

26. Della-Morte D, Dave KR, **DeFazio RA**, Bao YC, Raval AP, Perez-Pinzon MA: Resveratrol pretreatment protects rat brain from cerebral ischemic damage via a sirtuin 1-uncoupling protein 2 pathway. *Neuroscience*.159(3): 993-1002, 03/2009.PM19356683

27. Dave KR, Anthony Defazio R, Raval AP, Dashkin O, Saul I, Iceman KE, Perez-Pinzon MA, Drew KL: Protein kinase C epsilon activation delays neuronal depolarization during cardiac arrest in the euthermic arctic ground squirrel. *J Neurochem*.110(4): 1170-1179, 08/2009.PM19493168

28. Raval AP, Saul I, Dave KR, **DeFazio RA**, Perez-Pinzon MA, Bramlett H: Pretreatment with a single estradiol-17beta bolus activates cyclic-AMP response element binding protein and protects CA1 neurons against global cerebral ischemia. *Neuroscience*.160(2): 307-318, 05/2009.PM19272413

29. Lin B, Levy S, Raval AP, Perez-Pinzon MA, **Defazio RA**: Forebrain ischemia triggers GABAergic system degeneration in substantia nigra at chronic stages in rats. *Cardiovasc Psychiatry Neurol*.2010: 506952, 01/2010.PM20981346

30. Deniz E, Tomasulo M, **Defazio RA**, Watson BD, Raymo FM: Fluorescence patterning in films of a photoswitchable BODIPY-spiropyran dyad. *Phys Chem Chem Phys*.12(37): 11630-11634, 10/2010.PM20714479

31. Abaffy T, Duncan R, Riemer DD, Tietje O, Elgart G, Milikowski C, **DeFazio RA**: Differential volatile signatures from skin, naevi and melanoma: a novel approach to detect a pathological process. *PLoS One*.5(11): e13813, 11/2010.PM21079799

32. Lin HW, **Defazio RA**, Della-Morte D, Thompson JW, Narayanan SV, Raval AP, Saul I, Dave KR, Perez-Pinzon MA: Derangements of post-ischemic cerebral blood flow by protein kinase C delta. *Neuroscience*.171(2): 566-576, 12/2010.PM20813167

33. Abaffy T, Möller M, Riemer DD, Milikowski C, **Defazio RA**: A case report - Volatile metabolomic signature of malignant melanoma using matching skin as a control. *J Cancer Sci Ther*.3(6): 140-144, 01/2011.PM22229073

34. Dave KR, Bhattacharya SK, Saul I, **DeFazio RA**, Dezfulian C, Lin HW, Raval AP, Perez-Pinzon MA: Activation of protein kinase C delta following cerebral ischemia leads to release of cytochrome C from the mitochondria via bad pathway. *PLoS One*.6(7): e22057, 01/2011.PM21789211

35. **Defazio RA**, Levy S, Morales CL, Levy RV, Dave KR, Lin HW, Abaffy T, Watson BD, Perez-Pinzon MA, Ohanna V: A protocol for characterizing the impact of collateral flow after distal middle cerebral artery occlusion. *Transl Stroke Res*.2(1): 112-127, 03/2011.PM21593993

36. Abaffy T, Defazio AR: The location of olfactory receptors within olfactory epithelium is independent of odorant volatility and solubility. *BMC Res Notes*.4: 137, 05/2011.PM21548958

37. Pielecka-Fortuna J, **DeFazio RA**, Moenter SM: Voltage-gated potassium currents are targets of diurnal changes in estradiol feedback regulation and kisspeptin action on gonadotropin-releasing hormone neurons in mice. *Biol Reprod*.85(5): 987-995, 11/2011.PM21778142

38. Foster TC, **Defazio RA**, Bizon JL: Characterizing cognitive aging of spatial and contextual memory in animal models. *Front Aging Neurosci*.4: 12, 01/2012.PM22988436

39. Roberson ED, **Defazio RA**, Barnes CA, Alexander GE, Bizon JL, Bowers D, Foster TC, Glisky EL, Levin BE, Ryan L, Wright CB, Geldmacher DS: Challenges and opportunities for characterizing cognitive aging across species. *Front Aging Neurosci*.4: 6, 01/2012.PM22988434

40. Lin HW, Della-Morte D, Thompson JW, Gresia VL, Narayanan SV, **Defazio RA**, Raval AP, Saul I, Dave KR, Morris KC, Si M-L, Perez-Pinzon MA: Differential effects of delta and epsilon protein kinase C in modulation of postischemic cerebral blood flow. *Adv Exp Med Biol*.737: 63-69, 01/2012.PM22259083

41. Raval AP, Sick JT, Gonzalez GJ, **Defazio RA**, Dong C, Sick TJ: Chronic nicotine exposure inhibits estrogen-mediated synaptic functions in hippocampus of female rats. *Neurosci Lett*.517(1): 41-46, 05/2012.PM22521583

42. **Defazio RA**, Zhao W, Deng X, Obenaus A, Ginsberg MD: Albumin therapy enhances collateral perfusion after laser-induced middle cerebral artery branch occlusion: a laser speckle contrast flow study. *J Cereb Blood Flow Metab*.32(11): 2012-2022, 11/2012.PM22781334

43. Abaffy T, Möller MG, Riemer DD, Milikowski C, **DeFazio RA**: Comparative analysis of volatile metabolomics signals from melanoma and benign skin: a pilot study. *Metabolomics*.9(5): 998-1008, 01/2013.PM24039618

44. **DeFazio RA**, Elias CF, Moenter SM: GABAergic transmission to kisspeptin neurons is differentially regulated by time of day and estradiol in female mice. *J Neurosci*.34(49): 16296-16308, 12/2014.PM25471569

45. Wang L, **DeFazio RA**, Moenter SM: Excitability and Burst Generation of AVPV Kisspeptin Neurons Are Regulated by the Estrous Cycle Via Multiple Conductances Modulated by Estradiol Action. *eNeuro*.3(3)01/2016.PM27280155

46. Obenaus A, Ng M, Orantes AM, Kinney-Lang E, Rashid F, Hamer M, **DeFazio RA**, Tang J, Zhang JH, Pearce WJ: Traumatic brain injury results in acute rarefication of the vascular network. *Sci Rep*.7(1): 239, 03/2017.PM28331228

47. Vanacker C, Moya MR, DeFazio RA, Johnson ML, Moenter SM: Long-Term Recordings of Arcuate Nucleus Kisspeptin Neurons Reveal Patterned Activity That Is Modulated by Gonadal Steroids in Male Mice. *Endocrinology*.158(10): 3553-3564, 10/2017.PM28938398

48. Silveira MA, Burger LL, DeFazio RA, Wagenmaker ER, Moenter SM: GnRH Neuron Activity and Pituitary Response in Estradiol-Induced vs Proestrous Luteinizing Hormone Surges in Female Mice. *Endocrinology*.158(2): 356-366, 02/2017.PM27911605

49. Adams C, Stroberg W, **DeFazio RA**, Schnell S, Moenter SM: Gonadotropin-Releasing Hormone (GnRH) Neuron Excitability Is Regulated by Estradiol Feedback and Kisspeptin. *J Neurosci*.38(5): 1249-1263, 01/2018.PM29263236

50. Salehi A, Jullienne A, Wendel KM, Hamer M, Tang J, Zhang JH, Pearce WJ, **DeFazio RA**, Vexler ZS, Obenaus A: A Novel Technique for Visualizing and Analyzing the Cerebral Vasculature in Rodents. *Transl Stroke Res*.05/2018.PM29766452

51. Adams C, DeFazio RA, Christian CA, Milescu LS, Schnell S, Moenter SM: Changes in Both Neuron Intrinsic Properties and Neurotransmission Are Needed to Drive the Increase in GnRH Neuron Firing Rate during Estradiol-Positive Feedback. *J Neurosci*.39(11): 2091-2101, 03/2019.PM30655354

52. **DeFazio RA**, Navarro MA, Adams CE, Milescu LS, Moenter SM: Estradiol Enhances the Depolarizing Response to GABA and AMPA Synaptic Conductances in Arcuate Kisspeptin Neurons by Diminishing Voltage-Gated Potassium Currents. *J Neurosci*.39(48): 9532-9545, 11/2019.PM31628184

53. Dulka EA, DeFazio RA, Moenter SM: Chemogenetic Suppression of GnRH Neurons during Pubertal Development Can Alter Adult GnRH Neuron Firing Rate and Reproductive Parameters in Female Mice. *eNeuro*.7(3)01/2020.PM32513661

54. Penix J, DeFazio RA, Dulka EA, Schnell S, Moenter SM: Firing patterns of gonadotropin-releasing hormone neurons are sculpted by their biologic state. *R Soc Open Sci*.7(8): 201040, 08/2020.PM32968535

55. **DeFazio RA**, Moenter SM: Gonadotropin-Releasing Hormone (GnRH) Neuron Potassium Currents and Excitability in Both Sexes Exhibit Minimal Changes upon Removal of Negative Feedback. *eNeuro*.8(4)01/2021.PM34135001

56. Starrett JR, **DeFazio RA**, Moenter SM: Reciprocal Changes in Voltage-Gated Potassium and Subthreshold Inward Currents Help Maintain Firing Dynamics of AVPV Kisspeptin Neurons during the Estrous Cycle. *eNeuro*.8(5)01/2021.PM34385153

57. Vanacker C, Defazio RA, Sykes CM, Moenter SM: A role for glial fibrillary acidic protein (GFAP)-expressing cells in the regulation of gonadotropin-releasing hormone (GnRH) but not arcuate kisspeptin neuron output in male mice. *Elife*.1007/2021.PM34292152

58. Courtney CD, Sobieski C, Ramakrishnan C, Ingram RJ, Wojnowski NM, **DeFazio RA**, Deisseroth K, Christian-Hinman CA: Optoα1AR activation in astrocytes modulates basal hippocampal synaptic excitation and inhibition in a stimulation-specific manner. *Hippocampus*.33(12): 1277-1291, 12/2023.PM37767862

59. Jaime J, **DeFazio RA**, Moenter SM: Development and prenatal exposure to androgens alter potassium currents in gonadotropin-releasing hormone neurons from female mice. *J Neuroendocrinol*.36(3): e13373, 03/2024.PM38403894

60. Phillips CD, **DeFazio RA**, Moenter SM: Sex and Time of Day Alter the Interactions Between Hypothalamic Glia and the Neural Circuits Controlling Reproduction. *Endocrinology*.166(5)03/2025.PM40111184

61. Jaime J, **DeFazio RA**, Moenter SM: Response of gonadotropin-releasing hormone neurons from female mice to dynamic-clamp-simulated GABAergic conductances across development and after prenatal androgenization. *J Neuroendocrinol*.38(2): e70144, 02/2026.PM41652850

62. Chen X, Jaime J, **DeFazio RA**, Moenter SM: Voltage-gated calcium channel activity of gonadotropin-releasing hormone (GnRH) neurons is altered by age and by prenatal androgen exposure in female mice. *J Neuroendocrinol*.38(9): e70268, 09/2026. PM42759952. Preprint: bioRxiv 02/2026, PM42124562

Reviews

1. Moenter SM, DeFazio AR, Pitts GR, Nunemaker CS: Mechanisms underlying episodic gonadotropin-releasing hormone secretion. *Front Neuroendocrinol*.24(2): 79-93, 04/2003.PM12762999

2. Moenter SM, **DeFazio RA**, Straume M, Nunemaker CS: Steroid regulation of GnRH neurons. *Ann N Y Acad Sci*.1007: 143-152, 12/2003. PM14993048

3. Raval AP, Lin HW, Dave KR, **Defazio RA**, Della Morte D, Kim EJ, Perez-Pinzon MA: Resveratrol and ischemic preconditioning in the brain. *Curr Med Chem*.15(15): 1545-1551, 01/2008.PM18537630

Books and Chapters

1. Dudek FE, Pouliot WA, Kuehl-Kovarik C, **DeFazio RA**, Partin KM, Handa RJ, Moenter SM: Electrophysiological analyses of GnRH neurons using a transgenic mouse model. CRC Press.. In Handa RJ, Hayashi S, Terasawa E, Kawata M Eds. *NEUROPLASTICITY, DEVELOPMENT, AND STEROID HORMONE ACTION.*,CRC Press, (2002)
2. Perez-Pinzon MA, Dave KR, **DeFazio RA**, Raval AP, Kim EJ, Della Morte D: Neuroprotection: Endogenous Mechanisms. *Encyclopedia of Neuroscience*,(2009), 961-965, DOI:10.1016/B978-008045046-9.00517-9
3. Hughes S, Dashkin O, **DeFazio RA**: Vessel painting technique for visualizing the cerebral vascular architecture of the mouse. In Richard Milner *Cerebral Angiogenesis, Methods and Protocols. Methods in Molecular Biology 1135*,Humana Press, (2014)

In Press

1. **DeFazio RA**, Burgess CR, Burger L, Gibson AG, Moenter SM: Arcuate kisspeptin neurons display two types of calcium events: regulation by sex, gonads, and senktide. *J Neuroendocrinol*, in press.

Abstract/Posters

1. Hablitz JJ, **DeFazio T**: Excitability changes in freeze-induced neocortical microgyria, Netherlands, *Epilepsy Res*.32(1-2): 75-82, 1998. DOI:10.1016/s0920-1211(98)00041-2
2. Hablitz JJ, **DeFazio RA**: GABA-mediated depolarizing waves visualized with voltage sensitive dyes, *EPILEPSIA*.40: 20-20, 1999
3. **DeFazio RA**, Hablitz JJ: Voltage sensitive dyes reveal increased horizontal spread in the freeze-lesion model of neocortical microgyria, *EPILEPSIA*.40: 156-156, 1999
4. Hablitz JJ, **DeFazio RA**: Altered receptor subunit expression in rat neocortical malformations, United States, *Epilepsia*.41 Suppl 6: S82-S85, 2000. DOI:10.1111/j.1528-1157.2000.tb01562.x
5. **DeFazio RA**, Moenter SM: Estrogen negative feedback alters A-type potassium currents in murine gonadotrpin-releasing hormone (GNRH) neurons, Endocrine Society, 2001
6. **DeFazio RA**, Moenter SM: A-type potassium currents determine action potential latency in GnRH neurons, Society for Neuroscience, 2001
7. **DeFazio RA**, Sullivan SD, Moenter SM: GABA excites and inhibits GnRH neurons, Society for Neuroscience, 2002
8. Moenter SM, Sullivan SD, **DeFazio RA**: Endogenous GABA can excite GnRH neurons, Endocrine Society, 2004
9. **DeFazio RA**, Richter TA, Roper SD: Cold inhibits sour taste transduction, Association for Chemoreception Sciences (AChemS), 2004
10. **DeFazio RA**, Dvoriantchikov G, Roper SD, Chaudhari N: Single cell RT-PCR and functional characterization of mouse circumvallate taste cells, Association for Chemoreception Sciences, 2005
11. **DeFazio RA**, Dashkin O, Raval AP, Perez-Pinzon MA: Ischemic preconditioning increases GABA synaptic inputs to CA1 hippocampal neurons, Society for Neuroscience., 2006
12. **DeFazio RA**, Raval AP, Perez-Pinzon MA: Ischemic preconditioning confers neuroprotection via GABA synapses, Brain07, Osaka, Japan, 2007
13. **DeFazio RA**, Dashkin O, Perez-Pinzon MA, Dave KR, Lin HW, della Morte D, Saul I, Raval AP: Enhanced interneuron firing and network activity in CA1 region of the rat hippocampus after cardiac arrest, Society for Neuroscience, 2008

14. Abaffy T, **DeFazio RA**: Is there Functionality to the Spatial Distribution of Olfactory Receptors within Dorsal Zone of Olfactory Epithelium?, *CHEMICAL SENSES*.33(8): S151-S151, 2008

15. Maruyama Y, Huang Y-J, Dvoryanchikov G, Kim JW, **DeFazio RA**, Chaudhari N, Roper SD: Mechanisms for neurotransmission from mouse taste buds, *CHEMICAL SENSES*.33(2): J1-J2, 2008

16. Raval AP, Saul I, Dave KR, **Defazio RA**, Bramlett H, Perez-Pinzon MA: Estrogen priming protects hippocampus against cerebral ischemia in rat, *STROKE*.39(2): 666-666, 2008

17. Ginsberg MD, Park H-P, Nimmagadda A, **DeFazio RA**, Busto R, Prado R: Albumin treatment improves microvascular hemodynamics and augments the effect of thrombolytic therapy in arteriolar thrombosis: A two-photon microscopy study, *STROKE*.39(2): 562-562, 2008

18. **DeFazio RA**, McCarthy M, Nath A: Neuronal subtype specific responses to HIV Transactivator protein, Tat, International Symposium of NeuroVirology, 2009

19. **DeFazio RA**, Watson BD: The "Skimming Lesion" model of cortical photochemical stroke minimizes artifactual edema, Brain 09, Chicago, 2009

20. Hirsch N, Saul I, Dave K, **Defazio RA**, Bramlett H, Perez-Pinzon M, Raval A: Cyclic pattern of 17β-estradiol pretreatment protects the hippocampal ca1 region against cerebral ischemia, *JOURNAL OF CEREBRAL BLOOD FLOW AND METABOLISM*.29: S452-S453, 2009

21. Perez-Pinzon M, Dave K, Raval A, **DeFazio RA**, Saul I: Activation of protein kinase C delta following oxygen glucose deprivation leads to release of cytochrome c from mitochondria, *JOURNAL OF CEREBRAL BLOOD FLOW AND METABOLISM*.29: S164-S165, 2009

22. Abaffy T, Duncan R, Riemer D, Elgart G, Keri J, **DeFazio R**: P46 The volatile fingerprint of melanoma. *European Journal of Cancer Supplements*.7(4): 34, 10/2009. DOI:10.1016/s1359-6349(09)72221-9

23. **DeFazio RA**, Ohanna V, Levy S, Morales CL, Watson BD, Perez-Pinzon MA, Levy RV: Collateral flow after distal middle cerebral artery branch occlusion in the mouse, Society for Neuroscience Meeting, San Diego, CA, 2010

24. **DeFazio RA**, Deng X, McManus M, Perez-Pinzon MA, Lin HW, Silva L: Rapid remodeling of collaterals and neuroprotection after second ischemic stroke in aged mice, Society for Neuroscience, Washington DC, 2011

25. **DeFazio RA**, Elias C, Moenter SM: Chloride homeostasis and response to GABA in kisspeptin neurons of the AVPV and arcuate nucleus, Society for Neuroscience, New Orleans, 2012

26. **DeFazio RA**, Elias C, Moenter: Diurnal regulation of GABAergic synaptic inputs to anteroventral--periventricular (AVPV) area kisspeptin neurons, Society for Neuroscience, San Diego, CA, 2013

27. **DeFazio RA**, Elias C, Moenter SM: Estradiol modulates the number of GABAA receptors activated by GABA at arcuate nucleus kisspeptin (KNDy) neuron synapses, Society for Neuroscience, 2014

28. **DeFazio RA**, Navarro M, Milescu L, Moenter SM: Estradiol regulation of A-type potassium currents sculpts the membrane potential response to GABA in arcuate kisspeptin neurons, Society for Neuroscience, 2015

29. **DeFazio RA**, Moenter SM: Estradiol feedback shapes postsynaptic potentials in arcuate KNDy neurons via K+ and Ca2+ channels, International Congress of Neuroendocrinology, Toronto, 2018

30. Sobieski C, Wojnowski NM, Ramakrishnan C, **DeFazio RA**, Deisseroth K, Christian CA: Optogenetic stimulation of astrocytic Gq signaling differentially alters synaptic transmission in hippocampal CA1 and layer V of sensorimotor cortex, Society for Neuroscience, 2018

31. Ng MT, Orantes AM, Kinney□Lang E, Rashid F, Hamer M, **DeFazio RA**, Tang J, Zhang JH, Pearce WJ, Obenaus A: Poster 37: Temporal Evaluation of the Vascular Network after Traumatic Brain Injury. *PM&R*.10(9S1): s20-s21, 09/2018. DOI:10.1016/j.pmrj.2018.08.072